CSI 2132 Additional material

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Linear Hashing

- A dynamic hash file:

  grows and shrinks gracefully

- initially the hash file comprises M primary buckets numbered 0, 1, … M-1

- the hashing process is divided into several phases (phase 0, phase 1, phase 2, …). In phase j, records are hashed according to hash functions h_j(key) and h_{j+1}(key)

- $h_j(key) = key \bmod (2^j * M)$

phase 0: $h_0(key) = key \bmod (2^0 * M)$, $h_1(key) = key \bmod (2^1 * M)$
phase 1: $h_1(key) = key \bmod (2^1 * M)$, $h_2(key) = key \bmod (2^2 * M)$
phase 2: $h_2(key) = key \bmod (2^2 * M)$, $h_3(key) = key \bmod (2^3 * M)$
… …

Linear Hashing

- h_j(key) is used first; to split, use h_{j+1}(key)
- splitting a bucket means to redistribute the records into two buckets: the original one and a new one. In phase j, to determine which ones go into the original while the others go into the new one, we use h_{j+1}(key) = key mod 2^{j+1}*M to calculate their address.
- splitting buckets

  splitting occurs according to a specific rule such as

  - an overflow occurring, or
  - the load factor reaching a certain value, etc.
- a split pointer keeps track of which bucket to split next
- split pointer goes from 0 to 2^j*M - 1 during the jth phase, j= 0, 1, 2, … …

Linear Hashing

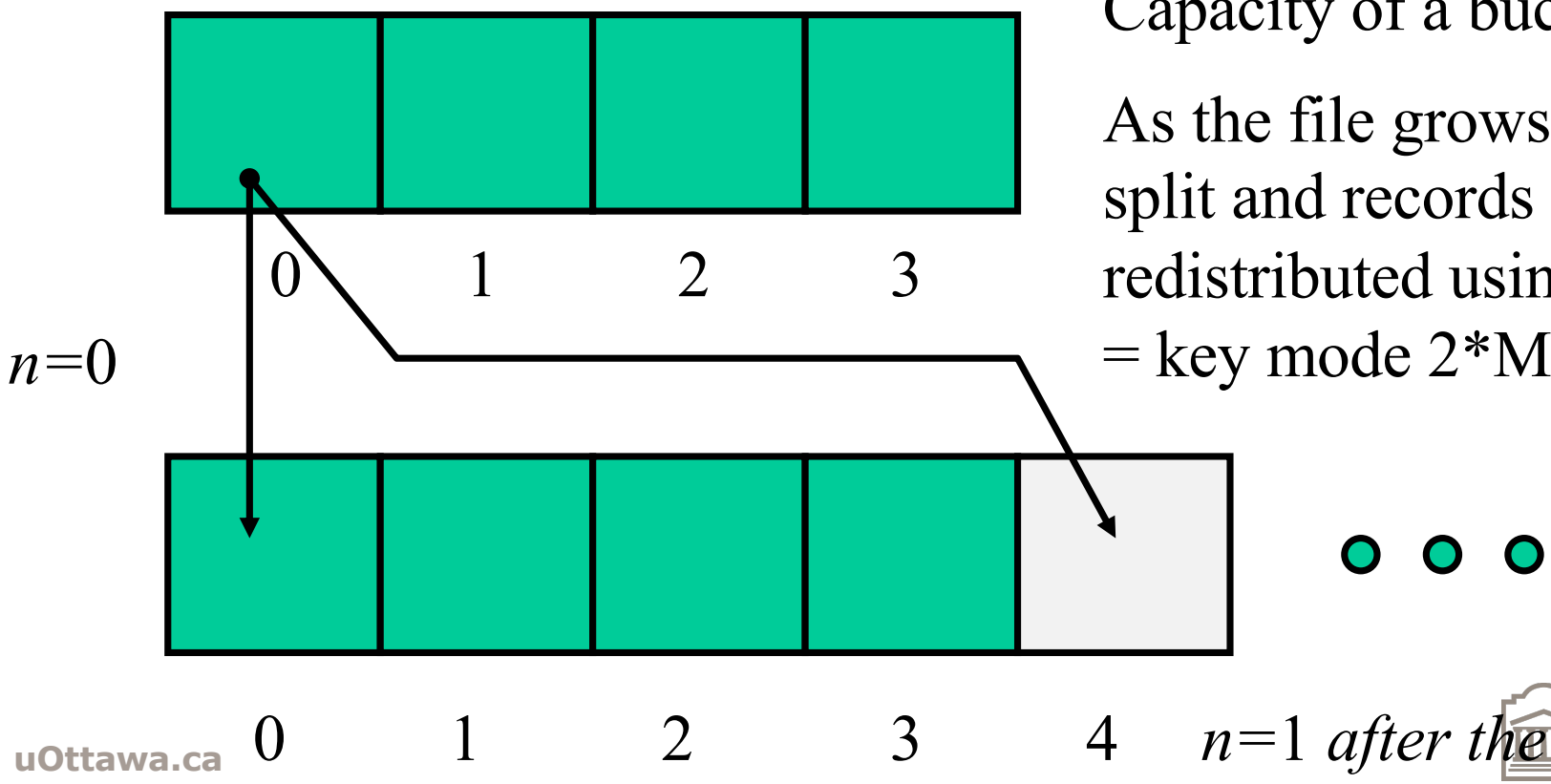
1. What is a phase?
2. When to split a bucket?
3. How to split a bucket?
4. What bucket will be chosen to split next?
5. How do we find a record inserted into a linear hashing file?

Linear Hashing, example

- initially suppose M=4
- h_0(key) = key mod M; i.e. key mod 4 (rightmost 2 bits)
- h_1(key) = key mod 2*M

Capacity of a bucket is 2.

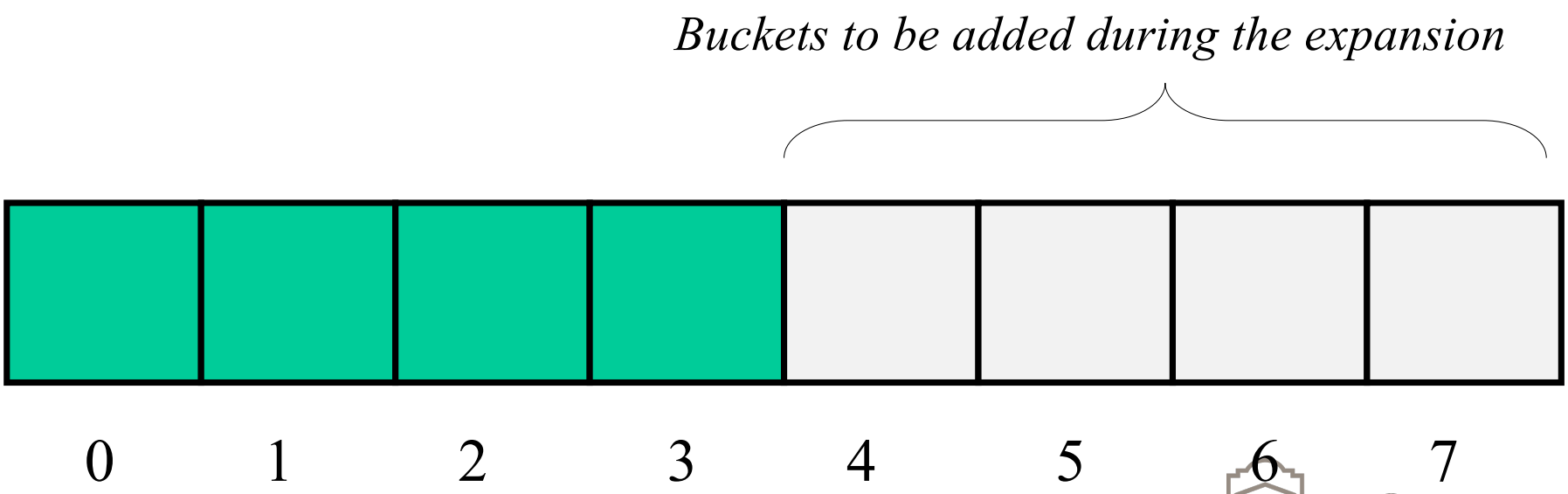
As the file grows, buckets split and records are redistributed using h_1(key) = key mode 2*M.



Linear Hashing, example

- split rule: if load factor > 0.70
- insert the records with key values:

0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100



Linear Hashing, example

- when inserting the sixth record (using h_0 = Key mod M) we would have

0100 1000	0001	0010 1110	0011
0	1	2	3

n=0 before the split

(n is the point to the bucket to be split.)

0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example

- when inserting the sixth record (using h_0 = Key mod M) we would have

0	1	2	3
0100 1000	0001	0010 1110	0011

n=0 before the split

(n is the point to the bucket to be split.)

- but the load factor 6/8= 0.75 > 0.70 and so bucket 0 must be split (using h_1 = Key mod 2M):

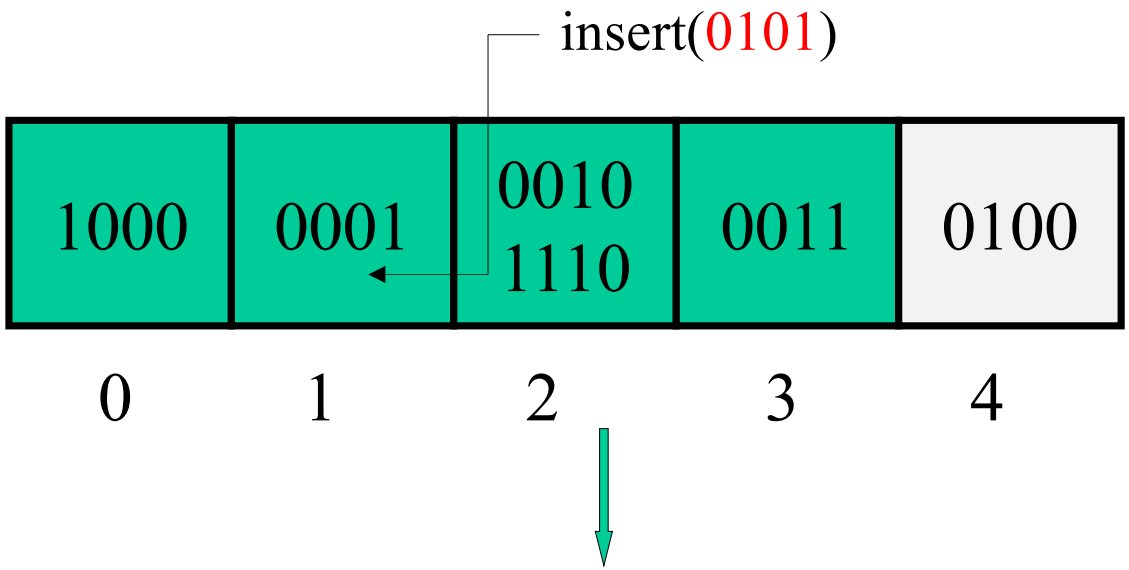
0	1	2	3	4
1000	0001	0010 1110	0011	0100

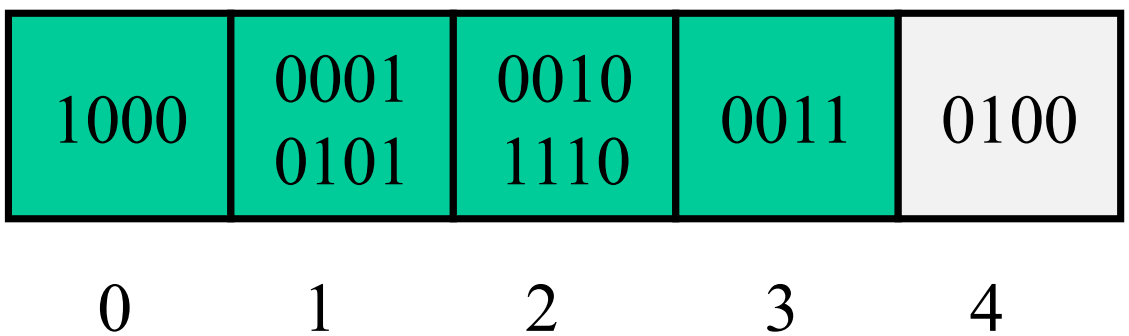
n=1 after the split

load factor: 6/10=0.6

no split

Linear Hashing, example





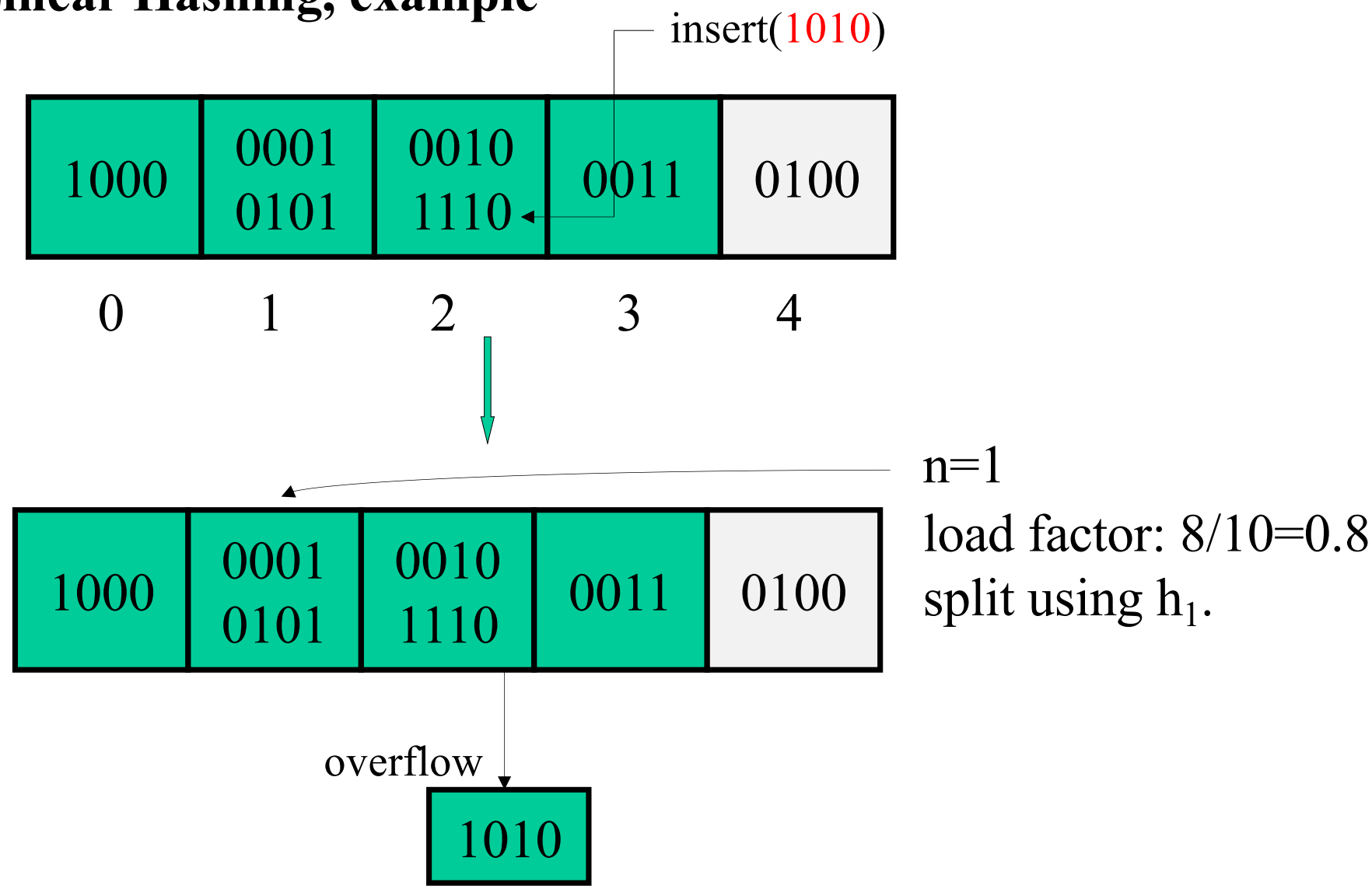
n=1

load factor: 7/10=0.7

no split

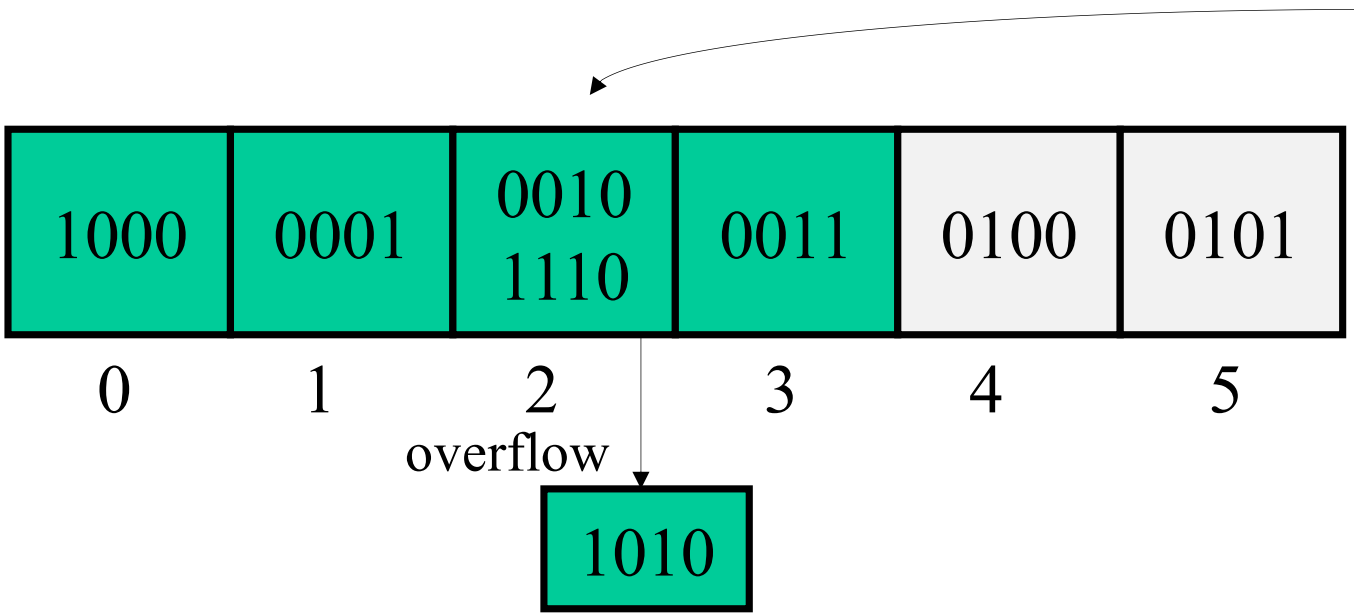
0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example



0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example



n=2

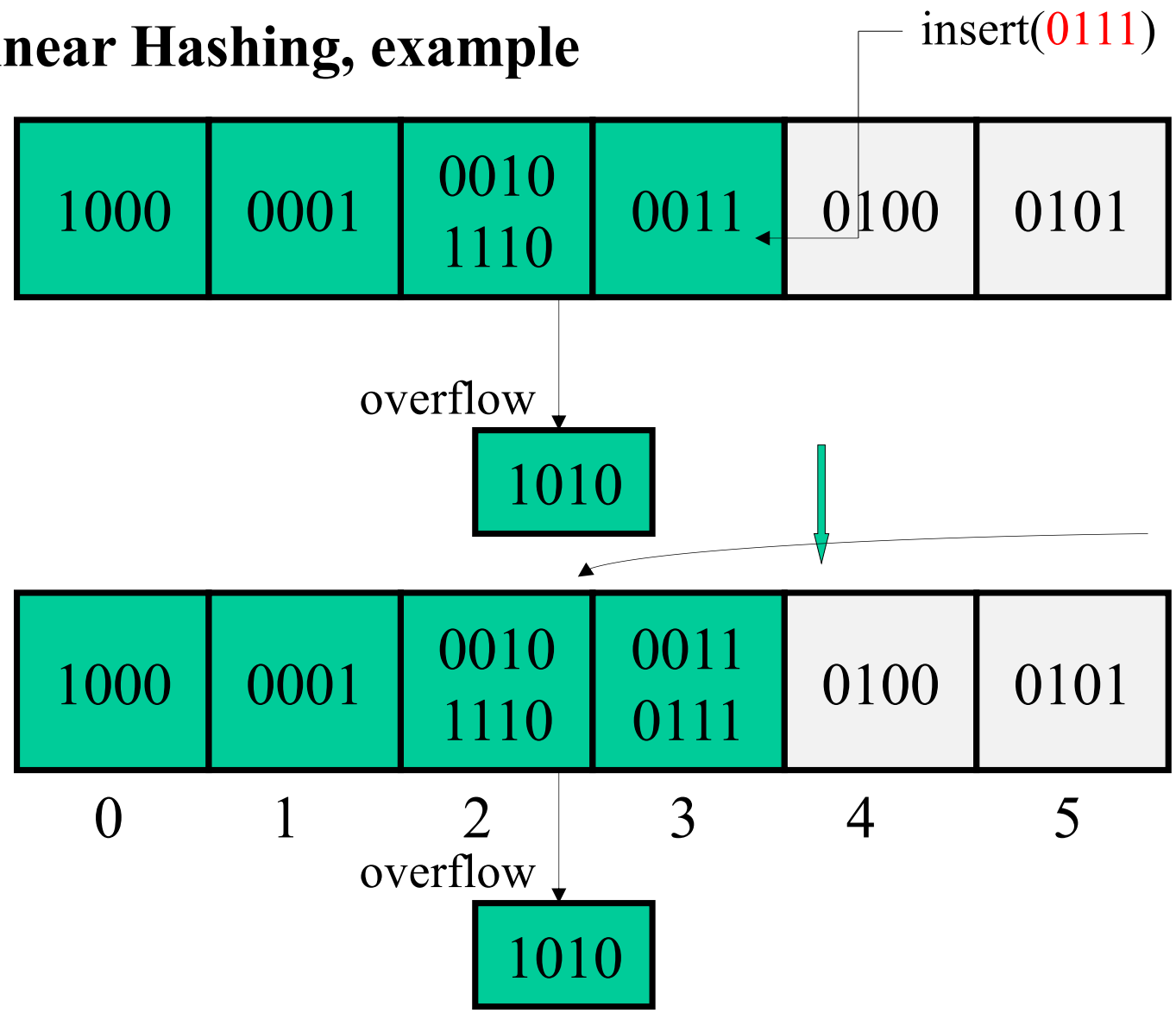
load factor:

8/12=0.66

no split

0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example



n=2

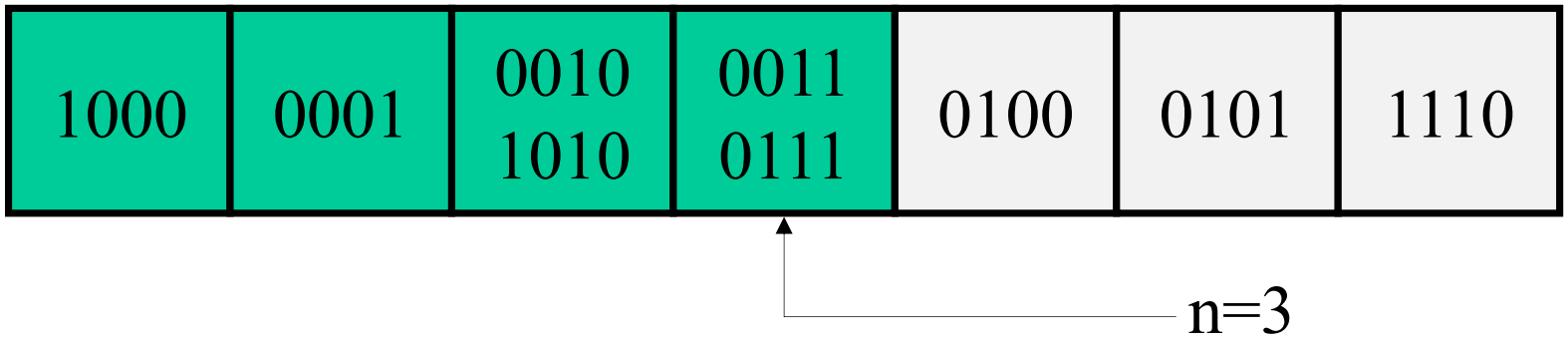
load factor:

9/12=0.75

split using h_1.

0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

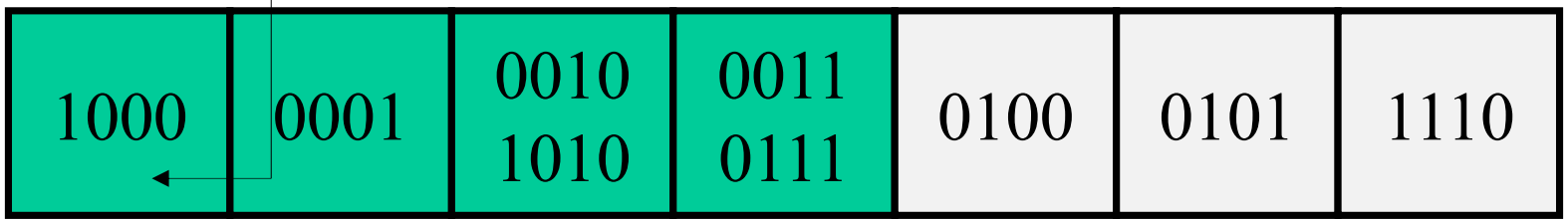
Linear Hashing, example



n=3

load factor: 9/14=0.642
no split.

insert(1100)



0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example

1000 1100	0001	0010 1010	0011 0111	0100	0101	1110

n=3

load factor: 10/14=0.71

split using h_1.

1000 1100	0001	0010 1010	0011	0100	0101	1110	0111

0011, 0010, 0100, 0001, 1000, 1110, 0101, 1010, 0111, 1100

Linear Hashing, example

1000 1100	0001	0010 1010	0011	0100	0101	1110	0111

n=4

load factor: 10/16=0.625

no split.

- At this point, all the 4 (M) buckets are split. n should be set to 0. It begins a second phase.
- In the second phase, we will use h_1 to insert records and h_2 to split a bucket.
  - note that $h_1(K) = K \bmod 2M$ and $h_2(K) = K \bmod 4M$.

Linear Hashing including two Phases:

- split rule: load factor > 0.7
- initially M = 4 (M: size of the primary area)
- hash functions: $h_i(\text{key}) = \text{key mod } 2^i \times M$ (i = 0, 1, 2, …)
- bucket capacity = 2

Trace the insertion process of the following keys into a linear hashing file:

3, 2, 4, 1, 8, 14, 5, 10, 7, 24, 17, 13, 15.

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010, 7 111, 24 11000, 17 10001, 13 1101, 15 1111

The first phase – phase$_0$

- when inserting the sixth record we would have

0	1	2	3
4 8	1	2 14	3

n=0 before the split

(n is the point to the bucket to be split.)

- but the load factor 6/8= 0.75 > 0.70 and so bucket 0 must be split (using h_1 = Key mod 2M):

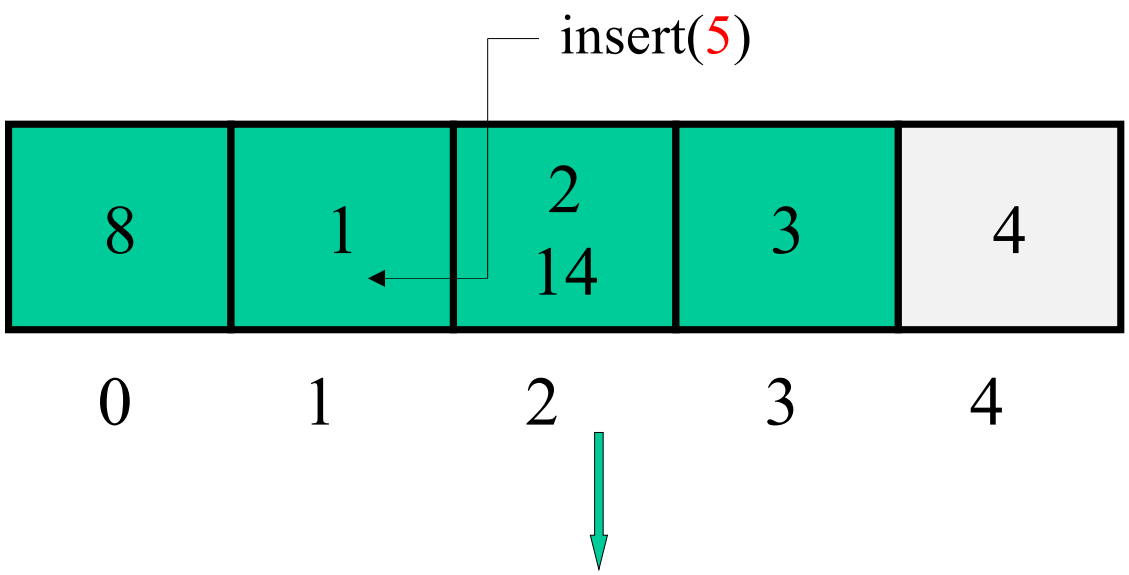
0	1	2	3	4
8	1	2 14	3	4

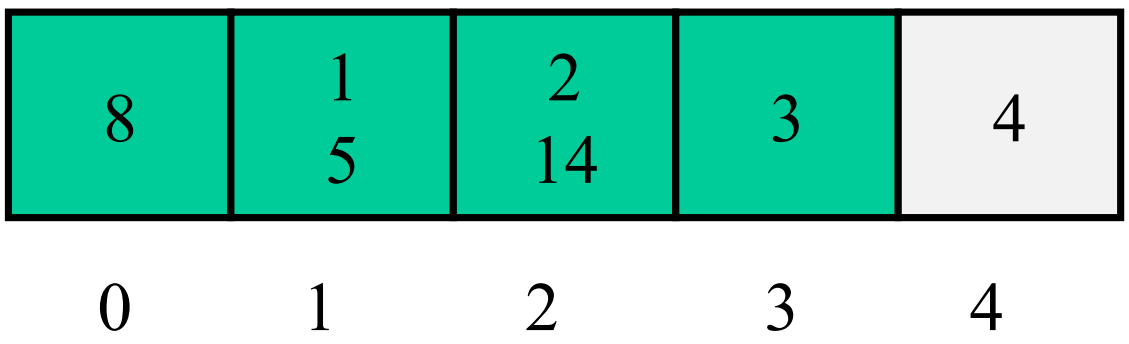
n=1 after the split

load factor: 6/10=0.6

no split

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010, 7 111, 24 11000, 17 10001, 13 1101, 15 1111



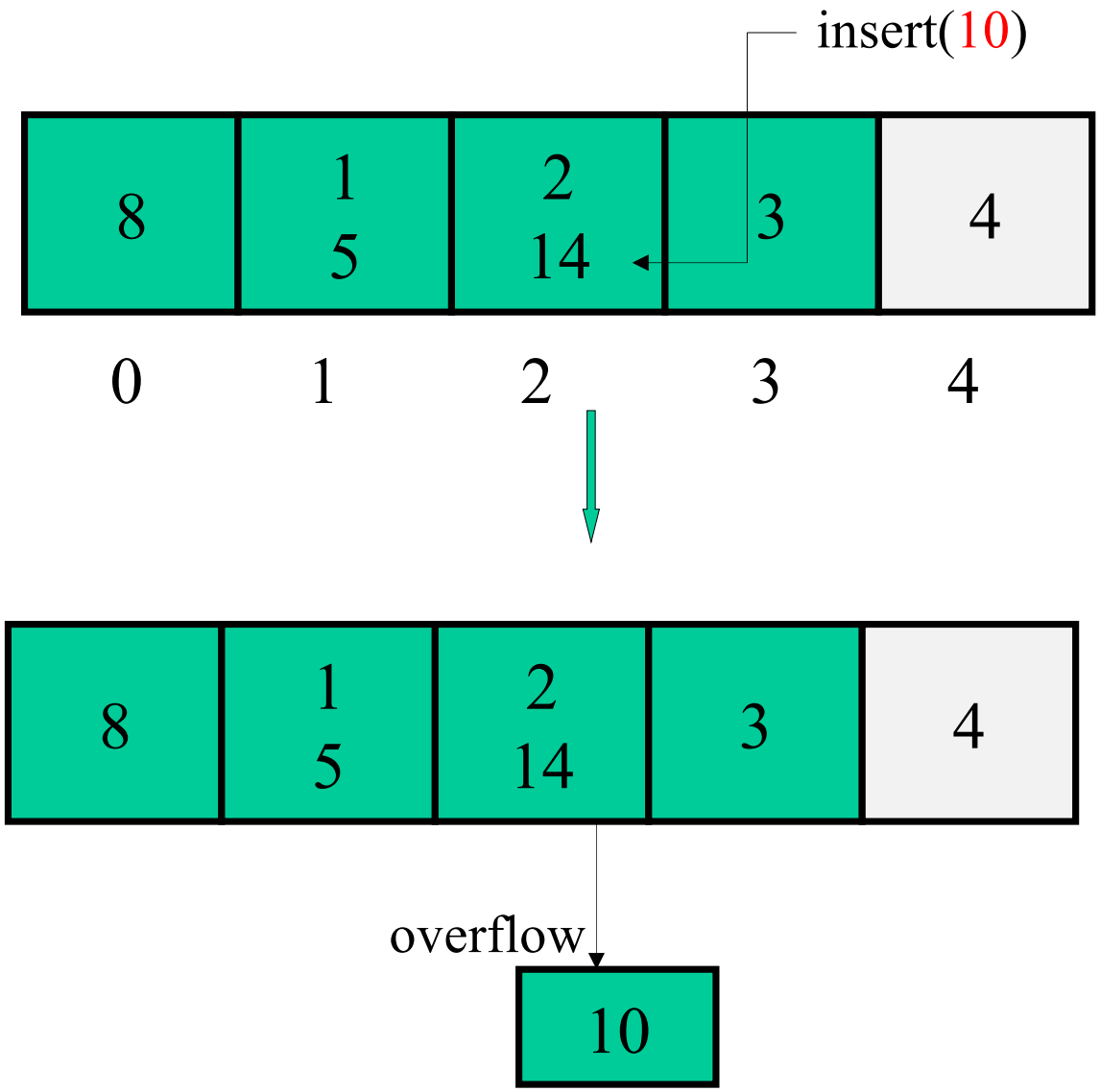


n=1

load factor: 7/10=0.7

no split

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

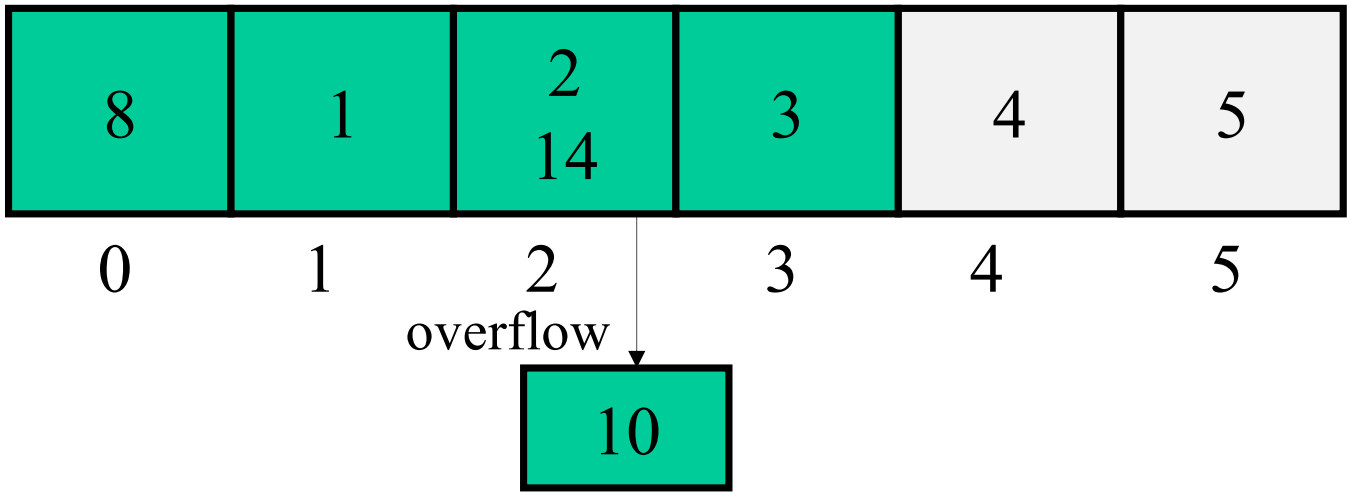


n=1

load factor: 8/10=0.8

split using h_1.

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

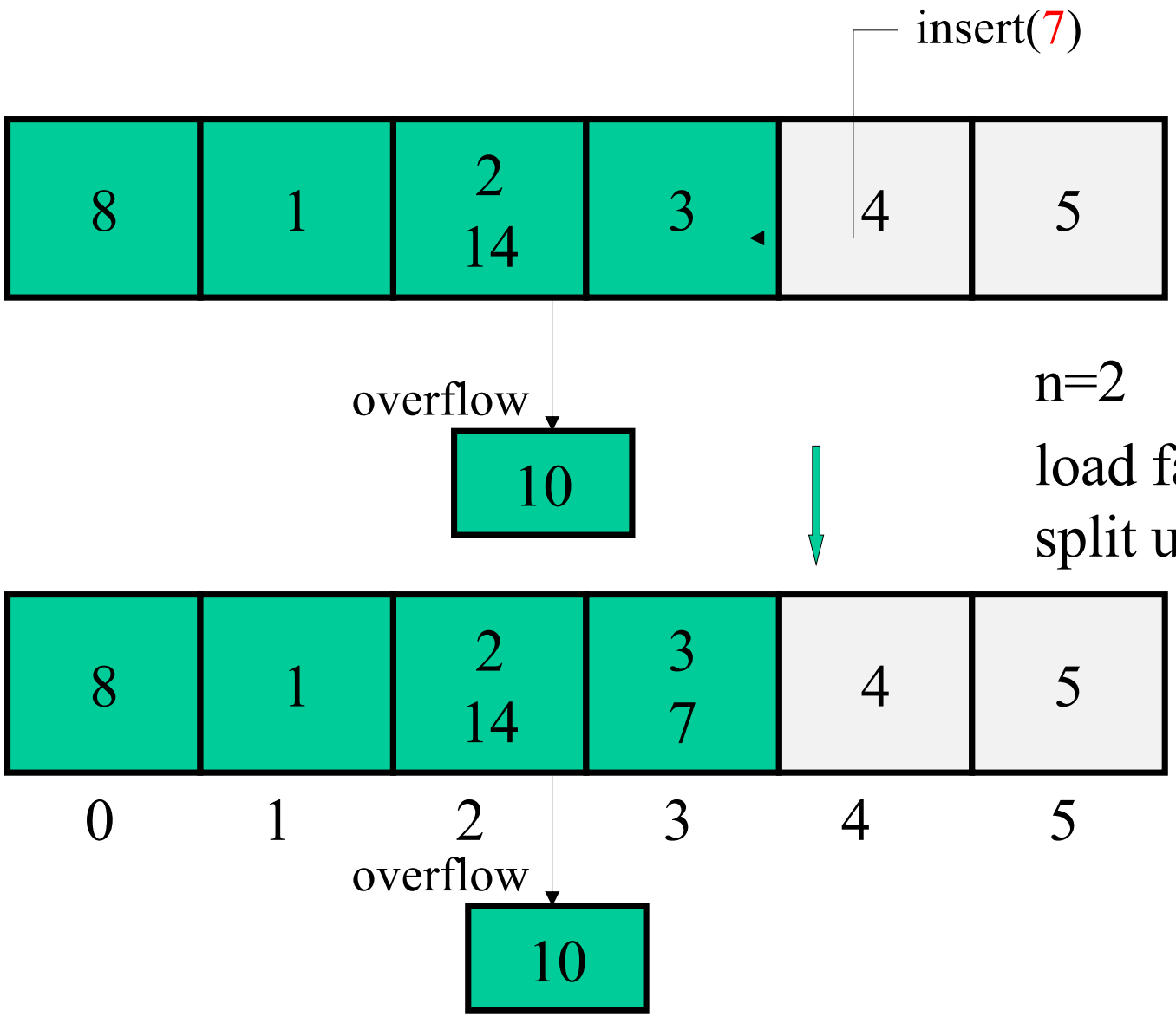


n=2

load factor: 8/12=0.66

no split

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111



n=2

load factor: 9/12=0.75

split using h_1.

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010, 7 111, 24 11000, 17 10001, 13 1101, 15 1111

8	1	2 10	3 7	4	5	14

n=3

load factor: 9/14=0.642

no split.

insert(24)

8	1	2 10	3 7	4	5	14

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

8 24	1	2 10	3 7	4	5	14

n=3

load factor: 10/14=0.71

split using h_1.

8 24	1	2 10	3	4	5	14	7

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

8 24	1	2 10	3	4	5	14	7

n=4

The second phase – phase$_1$

n = 0; using h_1 = Key mod 2M to insert and
h_2 = Key mod 4M to split.

insert(17)

8 24	1	2 10	3	4	5	14	7

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

8 24	1 17	2 10	3	4	5	14	7

n=0

load factor: 11/16=0.687

no split.

insert(13)

8 24	1 17	2 10	3	4	5	14	7

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010, 7 111, 24 11000, 17 10001, 13 1101, 15 1111

8 24	1 17	2 10	3	4	5 13	14	7

n=0

load factor: 12/16=0.75

split bucket 0, using h_2:

h_2 = Key mod 4M

	1 17	2 10	3	4	5 13	14	7	8 24

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010, 7 111, 24 11000, 17 10001, 13 1101, 15 1111

3 11, 2 10, 4 100, 1 01, 8 1000, 14 1110, 5 101, 10 1010,
7 111, 24 11000, 17 10001, 13 1101, 15 1111

insert(15)

	1 17	2 10	3	4	5 13	14	7	8 24

	1 17	2 10	3	4	5 13	14	7 15	8 24

n=1

load factor: 13/18=0.722
split bucket 1, using h_2.

	1 17	2 10	3	4	5 13	14	7 15	8 24	

Linear Hashing, example with splitting when there is overflow

Given have a **linear hash table** with:

- An **initial capacity** of M = 4 buckets: 0, 1, 2, 3.
- A **hash function**: h(K)= K mod M, where M = 4 initially.

Therefore:

- h_0(key) = key mod M; i.e. key mod 4 (rightmost 2 bits)
- h_1(key) = key mod 2*M

- The table **splits** when any bucket overflows, starting from B0.
- Each bucket can hold a maximum of 2 keys before overflowing.
- Splitting doubles the table size by dynamically adjusting M and rehashing certain keys.

Linear Hashing, example with splitting when there is overflow

Instructions:

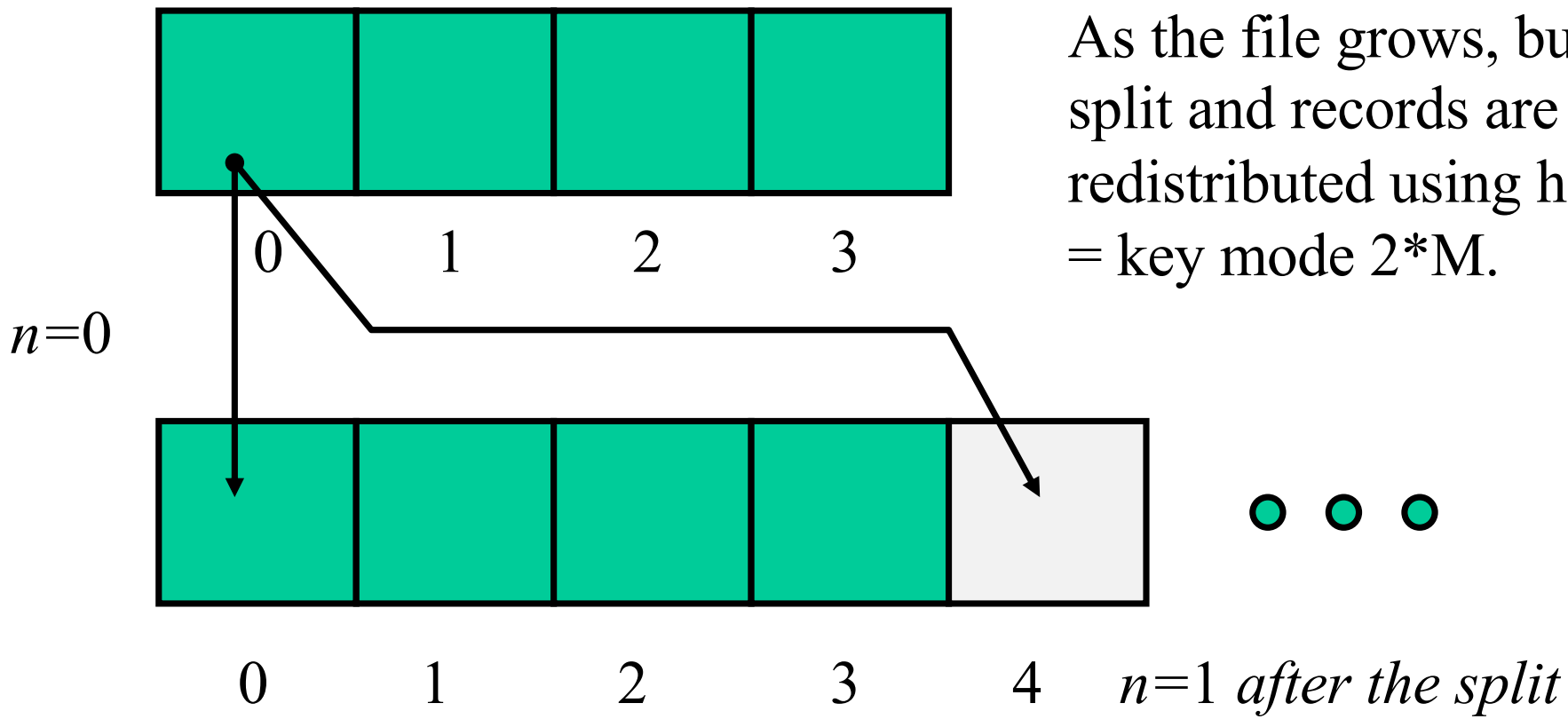
1. Insert the following keys in order: [5, 12, 7, 22, 3, 8, 14, 10].
2. Perform splitting **only** when a bucket overflows.
3. Show how the hash table evolves at each step, including:
    1. Which bucket is split.
    2. The updated values of M (modulus).
    3. Rehashed keys if necessary.

Linear Hashing, example with splitting when there is overflow

Solution:

Capacity of a bucket is 2.

As the file grows, buckets split and records are redistributed using h_1(key) = key mode 2*M.



Linear Hashing, example with splitting when there is overflow

n=0

0	1	2	3
12 8	5	22 14	7 3

h(5) = 5 mod 4 = 1. Key 5 is inserted into 1.
h(12) = 12 mod 4 = 0. Key 12 is inserted into 0.
h(7) = 7 mod 4 = 3. Key 7 is inserted into 3.
h(22) = 22 mod 4 = 2. Key 22 is inserted into 2.
h(3) = 3 mod 4 = 3. Key is inserted into 3.
h(8) = 8 mod 4=0. Key 8 is inserted into 0.
h(14) = 14 mod 4 = 2. Key is inserted into 2.

Linear Hashing, example with splitting when there is overflow

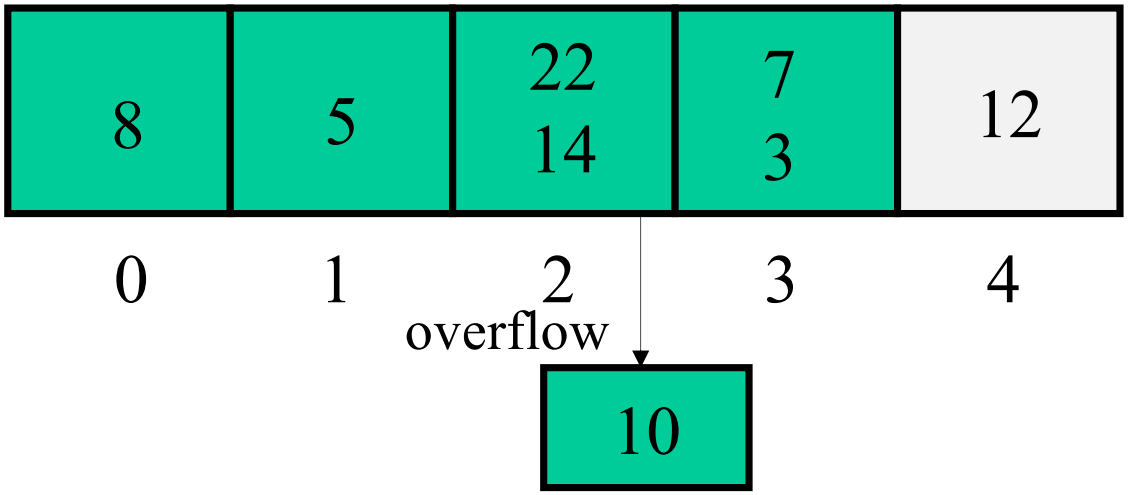
n=0

0	1	2	3
12 8	5	22 14	7 3

h(5) = 5 mod 4 = 1. Key 5 is inserted into 1.
h(12) = 12 mod 4 = 0. Key 12 is inserted into 0.
h(7) = 7 mod 4 = 3. Key 7 is inserted into 3.
h(22) = 22 mod 4 = 2. Key 22 is inserted into 2.
h(3) = 3 mod 4 = 3. Key is inserted into 3.
h(8) = 8 mod 4=0. Key 8 is inserted into 0.
h(14) = 14 mod 4 = 2. Key is inserted into 2.

Linear Hashing, example with splitting when there is overflow

n=1



h(10) = 10 mod 4 = 2.
Key 10 goes into 2, causing 2 to overflow.
At this point, the splitting process is triggered.
Bucket 0 is split to 0 and 4. 12 is rehashed to 4.